

YICHUN HEILONGJIANG, China

WMO: 507740

Lat: 47.7081N

Lon: 128.8358E

Elev: 850

StdP: 14.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-23.5	-32.6	0.9	-27.2	-28.7	1.1	-22.9	16.9	3.9	14.8	3.5	1.0	0	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	86.3	69.5	83.6	68.6	80.9	67.6	73.6	81.2	71.9	79.5	70.3	77.4	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	118.8	77.6	69.4	111.6	75.9	67.9	105.7	74.2	37.9	81.4	36.4	79.7	34.9	77.6	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.3	12.2	DB	-32.4	91.6	4.3	3.1	-35.5	93.9	-38.0	95.7	-40.4	97.5	-43.5	99.8	(4)
			WB	-32.6	76.0	4.2	1.8	-35.7	77.3	-38.2	78.3	-40.6	79.3	-43.6	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	-5.4	4.1	21.0	40.4	53.8	64.5	70.5	66.9	55.0	38.9	16.4	-2.0	(6)
(7)		DBStd	27.80	8.06	9.43	11.07	8.25	7.27	6.05	4.16	4.96	6.86	8.50	12.06	8.87	(7)
(8)		HDD50	7256	1719	1285	899	308	40	0	0	29	356	1007	1613		(8)
(9)		HDD65	11118	2184	1705	1364	740	354	80	6	36	306	808	1457	2078	(9)
(10)		CDD50	1964	0	0	0	18	159	436	635	524	179	13	0	0	(10)
(11)		CDD65	352	0	0	0	1	8	66	176	95	6	0	0	0	(11)
(12)		CDH74	3059	0	0	0	11	217	779	1314	661	74	3	0	0	(12)
(13)		CDH80	768	0	0	0	1	40	238	360	121	8	0	0	0	(13)
(14)	Wind	WSAvg	4.4	3.6	4.1	5.0	5.8	5.3	4.3	3.9	3.7	3.9	4.8	4.4	3.8	(14)
(15)	Precipitation	PrecAvg	25.3	0.2	0.2	0.4	0.8	2.1	3.9	6.5	6.4	2.9	1.1	0.5	0.3	(15)
(16)		PrecMax	39.2	0.4	0.6	0.9	2.5	5.2	8.1	13.6	13.4	5.1	3.0	1.3	0.9	(16)
(17)		PrecMin	17.9	0.0	0.0	0.1	0.1	0.6	0.8	2.4	0.9	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	5.1	0.1	0.2	0.2	0.5	1.2	1.5	2.8	2.9	1.2	0.8	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	37.5	54.2	74.5	84.5	91.3	90.4	86.4	80.7	70.9	52.2	32.7	(19)
MCWB			19.2	29.9	41.4	52.1	60.7	68.9	72.2	72.0	66.2	55.7	43.0	28.3	(20)	
2%		DB	16.0	31.6	47.9	67.7	79.9	85.7	86.8	83.5	75.6	63.4	45.0	21.0	(21)	
		MCWB	13.4	26.1	36.9	48.9	58.6	66.8	70.9	70.8	62.6	50.9	37.7	18.7	(22)	
5%		DB	12.3	26.4	42.9	62.4	75.6	82.4	84.0	80.7	72.0	58.3	38.9	16.5	(23)	
		MCWB	10.2	22.3	34.1	46.5	57.0	65.5	70.6	69.3	60.6	47.9	33.8	14.6	(24)	
10%		DB	8.7	21.2	38.4	57.2	70.7	78.4	81.2	77.9	68.3	53.8	34.1	12.6	(25)	
		MCWB	7.0	17.7	31.2	43.5	54.2	64.5	69.8	67.9	58.5	44.9	30.0	10.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	30.3	42.2	54.1	63.2	72.1	75.9	75.5	68.7	57.2	43.9	28.7	(27)
MDCB			22.0	36.6	53.0	70.4	77.9	84.7	83.4	82.8	77.4	67.4	50.5	32.3	(28)	
2%		WB	13.5	26.2	38.0	50.3	60.7	69.6	74.1	73.1	65.3	52.6	38.0	18.8	(29)	
		MDCB	15.7	31.3	46.4	65.7	75.2	81.0	81.9	79.9	71.6	60.7	44.1	20.8	(30)	
5%		WB	10.1	22.1	34.6	47.3	58.4	67.8	72.7	71.3	62.6	49.1	34.0	14.5	(31)	
		MDCB	12.2	26.4	41.9	60.5	71.9	78.1	80.4	77.6	68.9	57.0	38.4	16.4	(32)	
10%		WB	7.0	17.7	31.7	44.4	55.9	65.9	71.3	69.5	60.3	45.8	30.0	10.9	(33)	
		MDCB	8.7	21.1	37.7	55.9	68.0	74.9	78.8	75.5	66.4	52.7	34.0	12.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.5	26.6	23.2	22.5	23.9	21.5	18.3	18.4	22.8	20.8	19.8	20.3	(35)
MCDBR			24.8	28.7	25.6	31.5	33.0	29.5	24.3	23.2	28.5	29.6	24.0	21.4	(36)	
MCWBR			22.3	23.8	16.7	17.0	15.1	12.4	10.6	11.3	15.1	17.3	17.7	19.3	(37)	
5% WB		MCDBR	24.6	28.2	23.9	28.5	27.9	23.2	19.5	18.9	21.9	26.4	22.8	20.8	(38)	
		MCWBR	22.3	23.5	15.9	16.2	13.7	10.9	9.9	10.1	12.6	16.7	17.2	18.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.322	0.392	0.493	0.545	0.534	0.514	0.457	0.421	0.412	0.343	0.301	(40)	
(41)		taud	2.163	2.083	1.959	1.842	1.792	1.916	2.024	2.179	2.182	2.061	2.157	2.142	(41)	
(42)		Ebn at Noon	240	259	255	240	233	237	240	249	246	221	217	219	(42)	
(43)		Edh at Noon	27	36	49	61	66	59	52	43	39	37	27	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	609	960	1372	1512	1640	1712	1604	1484	1240	851	573	489	(44)	
(45)		RadStd	29	38	106	161	158	233	170	111	121	65	44	25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page